Case 1

Done	
Status	Cases

1. Psychiatric diagnosis: aims and what makes it different

Psychiatric diagnosis follows the same logic as medical diagnosis (history → hypotheses → signs → provisional diagnosis), but it rests mainly on the patient's subjective account, because no laboratory test can confirm a psychiatric disorder.

- **Symptom** = subjective, reported by the patient; **sign** = objective, observed by the psychiatrist
- Three aims of psychiatric diagnosis:
  1. Identify signs and symptoms → basis for **classifying** the disorder
  2. Track down **aetiological factors** → hypotheses about aetiopathogenesis
     - physical causes, traumatic life events
     - vulnerability: hereditary predisposition, life story, personality traits, coping styles, defence mechanisms
  3. Determine **severity and effects** (impairments, disabilities) → treatment is geared to these
- Differences from 'physical' diagnostics:
  - Emphasis on subjective symptoms → history-taking dominates
  - History-taking and examination are combined, not sequential
  - History is more detailed and includes social and biographical history → to find aetiological factors
  - Few lab tests → disorder and aetiology determined mainly from history + examination

- **Most fundamental:** instrument and object overlap
  - the functions the patient uses to report (thought, speech) may be the impaired functions under study
  - the psychiatrist's tools are his own mental functions (listening, looking, asking)
  - → a symptom can impair both the patient's report and the patient–psychiatrist interaction

When to suspect a physical cause

Almost every psychiatric disorder can be caused by a physical condition, so physical causes are ruled out before a disorder is called **primary**.

- Neurocognitive signs → physical cause is obvious
- Other disorders → the history, not typical neurocognitive signs, should raise suspicion
- Key rule: onset **after 40** with no previous episode → rule out physical causes carefully
  - because primary psychiatric disorders usually start long before that age
- Confirm suspicion with a brief neuropsychiatric screen; each test probes one function:
  - repeat a telephone number → attention
  - recall three words after minutes → short-term memory
  - similarity rose/tulip, meaning of sayings → abstraction
  - name > 10 animals in a minute → executive functions
  - draw a clock → visual agnosia / constructive apraxia
- If a physical cause is considered but unclear → lab screen for the commonest causes (blood count, liver, kidney, glucose, thyroid, vitamin B12/folate)

Psychiatric patients presenting with physical symptoms

Many psychiatric disorders present to the GP as physical complaints, and a large share go unrecognised.

- Suggestive clues:
  - complaint precipitated, perpetuated or worsened by psychosocial factors
  - unexplained symptoms fitting **depression** (fatigue, low energy, sleep, weight, low libido)
  - unexplained **vegetative/autonomic** symptoms fitting an **anxiety disorder** (palpitations, sweating, trembling, breathlessness, dizziness…)
  - excessive worry about serious illness
  - excessive alcohol, painkillers, benzodiazepines, drugs; medication with psychiatric side effects
  - personal or family psychiatric history (**previous episodes = strongest predictor of new ones**)
- Brief screen in every review of systems: sadness, anxiety, panic, phobias, illness anxiety, heavy drinking

2. History-taking

History-taking works outward from the chief complaint: each further layer either screens for other disorders (differential diagnosis) or uncovers causes, effects and context (aetiology, treatment).

Specific history (the chief complaint)

- = history of the disorder the patient is examined for
- Mental state examination starts **at the same time** (first impressions, how the patient communicates) → history and examination merge
- Interviewing stance:
  - start open-ended; look and listen before taking notes or asking specifics
  - don't assume you understand too quickly; avoid cross-examination
  - encourage difficult topics (suicidal or 'crazy' thoughts, sex), but cut through the 'psychological skin' only when necessary
  - show empathy; notice the emotions the patient evokes **in you**

- Dimensions of the chief complaint (same as for a physical complaint):
  - **Quality** – precise nature, what it resembles
  - **Location** – physical symptoms only
  - **Chronology** – onset, earlier episodes, course, frequency
  - **Quantity** – severity now and recently; effect on functioning (an average day)
  - **Setting** – where and when it occurs
  - **Influential factors** – what helps/worsens; self-help; treatment so far and its result
  - **Accompanying symptoms**

General psychiatric history = 'psychiatric review of systems'

- Purpose: differential diagnosis + guiding the mental state examination
- Screen key symptoms of the other main disorders with **sifting / detection questions**, grouped by cognitive, affective and conative functions
- Positive answer, **or** doubt despite a negative answer → explore in detail

Past psychiatric and family history

- Previous episodes and disorders in the family → differential diagnosis
- Response to previous treatments → clues for treatment

Physical history and examination

- Aim: find physical illness that (a) **causes** the disorder, (b) **results from** it, or (c) must be **considered in treatment**

Social history – three milieus

- First milieu = home (living situation, partner/children, social support, self-care, finances)
- Second milieu = work or education (role, performance, satisfaction, fit with abilities; if unemployed: why)

- Third milieu = wider roles (family of origin ties, civic participation, friends, brushes with the law)
- Each: the actual situation **and** how the patient perceives it
- Aims: (1) symptoms in daily life, (2) social causes and effects, (3) the social support system

Biographical history

- Family of origin, life history (development, relationships, education, career, psychotraumatic events), self-description
- Aims:
  - past psychiatric symptoms
  - predisposing and **protective** factors
  - factors significant for this disorder in the patient's **own narrative**
  - development of personality traits and self-image

History through an informant

- More often needed than in general medicine → perception of mental symptoms is partly subjective
- **Indispensable** with cognitive symptoms (delirium, dementia, psychosis) and complex, treatment-resistant disorders
- Functions:
  - verify, complement and correct the patient's account
  - 'objective' view of executive functioning the patient may not notice is failing (travel, cooking)
  - topics the patient avoids (substance use, antisocial or sexual behaviour)
  - the burden on those close to the patient

3. Examination: the mental state

Signs are established during history-taking by three techniques – exploration, observation, testing – and recorded in a mental state report organised by the 'mental triad' of thought, feeling and will (cognitive, affective, conative functions).

- Three techniques:
  - **Exploration** – targeted questions about symptoms suggested by what the patient reports
  - **Observation** – objective signs: appearance, behaviour, what the speech reveals; the only technique if the patient refuses to talk
  - **Testing** – specific test questions to establish and broadly quantify a sign; only when history/observation warrant it
- Mental triad and its functions:
  - **Cognitive** (thought): consciousness/attention/orientation · memory · intellectual functions (judgment, sense of reality and insight, intelligence/abstraction, language, metacognition) · imagination, perception, self-perception · thought form and content
  - **Affective** (feeling): mood · affect (emotion as displayed)
  - **Conative** (will): praxis and executive functions · psychomotor function · motivation, will and behaviour
- Report order: **first impressions** (appearance, contact, how symptoms are reported, feelings evoked in the examiner) → triad functions → **personality traits**
  - first impressions give clues to nature, severity and above all treatability

The hierarchy of syndromes

Psychiatric syndromes are ranked by urgency and required expertise; the ranking organises the report, the examination and classification.

1. **Neurocognitive disorders** (incl. substance-induced) – consciousness, attention, orientation, memory, intellect, perception → most urgent, most expertise
2. **Psychosis spectrum** – perception, thought form and content, psychomotor function, motivation

3. **Emotional disorders** – thought content, mood and affect, motivation
4. **Behavioural disorders** – motivation, will and behaviour → least urgent

- Pattern: moving down the hierarchy, the involved functions shift from cognitive → affective → conative
- The hierarchy sets the order and depth of the examination:
  - possible delirium, dementia, psychosis → mainly cognitive functions
  - panic attacks → start with affective symptoms and physical signs; cognitive/conative briefly
  - somatic symptom disorder → only broadly rule out cognitive and conative signs

Key distinctions in sign terminology

- Consciousness: somnolent (dozes, answers when spoken to forcefully) → soporific (no answer, but does simple tasks); **narrowed** = closed off, focused on one experience
- Amnesia:
  - **anterograde** – no memory for the period **after** an acute cerebral disorder
  - **retrograde** – no memory for the period **before** it
  - **dissociative** – cannot recall psychotraumatic personal experiences
  - **confabulation** – unconsciously fills amnestic gaps with made-up facts
- Insight: **illness awareness** = aware of the symptoms; **understanding of illness** = also sees they are pathological and need treatment
- Metacognition: impaired theory of mind (link behaviour ↔ others' thoughts, feelings, motives) · empathic ability · mentalising (behaviour arises from mental states)
- Perception:
  - **hallucination** – perceives something that is not there
  - **illusion** – briefly misperceives a real object, usually under emotion

    - **derealisation** (environment unreal) vs **depersonalisation** (self unreal)
- Thought form:
    - racing thoughts – quick jumps, still intelligible
    - tangentiality – answers unrelated to the question
    - derailment – narrative interrupted by irrelevant, incomprehensible remarks
    - incoherence – illogical, disconnected speech
    - thought blocking – abrupt stop in the train of thought
    - rumination (pondering a problem without solution) vs worrying (emotional weight out of proportion)
- Thought content:
    - **delusion** – incorrect belief that cannot be corrected
    - **overvalued idea** – dominated by irrational ideas
    - **compulsive thought** – intrusive, unwanted
    - **urge-driven thought** – about exciting activities, controllable only briefly
- Affect: **labile** (rapid shifts, no external cause) · **flat** (hardly any emotion) · **inadequate** (inconsistent with circumstances/thoughts)
- Catatonic-type signs: stupor (motionless, mute, fully conscious) · **posturing** (adopts unusual posture spontaneously) vs **catalepsy** (holds posture induced by examiner) · waxy flexibility · negativism · echo phenomena
- Behaviour:
    - **compulsive** – repeated pointless acts (checking)
    - **urge-driven** – acting on urges (bingeing, stealing, gambling, hair-pulling)
    - **impulsive** – suicide attempt, self-harm, rage outbursts
    - **addictive** – tolerance, loss of control, withdrawal
- Personality traits (DSM-5 clusters):
    - A – paranoid, schizoid, schizotypal (suspicious, aloof, eccentric)

- B – antisocial, borderline, histrionic, narcissistic (impulsive, dramatic, grandiose)
- C – avoidant, dependent, compulsive (inhibited, submissive, rigid)

Other examination tools

- **Clinimetrics**
  - structured interviews (high inter-rater reliability; built on the sifting questions) → reliable **classification**, mainly in research
    - CIDI – usable by trained research nurses → epidemiology
    - SCID – clinicians → clinical research
  - severity questionnaires → quantify severity and treatment effect once diagnosed
- **Psychotherapeutic assessment** – how the patient copes with difficulties and where they arose
  - aim = the patient's **perception** and meaning, not just the facts
  - maps vulnerabilities **and** strengths (coping, defences, use of support, ability to form a working relationship)
  - tools: open questions, reflection, others' reactions, homework (symptom diary)
- **Psychodiagnosis** – standardised psychological tests (IQ, personality, neuropsychological) → subtle brain disorders; capacity for intensive psychotherapy
- **Lab tests** – confirm no psychiatric diagnosis; used to find physical causes and monitor drugs (blood levels; e.g. thyroid and renal function on lithium)
- **EEG** – rule out epilepsy vs conversion/pseudoepileptic seizures, dissociation, narcolepsy
- **Brain imaging**
  - MRI → structure (infarcts, atrophy, tumours, MS lesions)

    - PET/SPECT → metabolism, transmitter synthesis, receptor density and occupancy
    - fMRI → blood oxygenation as index of regional activity
    - functional imaging = research tools, **not** routine diagnostics

4. Classification vs diagnosis

Psychiatry diagnoses at two levels: **classification** gives a standardisable, descriptive label for the syndrome ('what the problem is'); **diagnosis** gives an individual, integrated explanation of the patient's state that guides treatment.

Classification

- Symptoms are grouped into **syndromes** = symptoms that often occur together
    - **key symptoms** – linked to a disorder with some certainty (depression: sadness, anhedonia)
    - **optional symptoms** – complete the picture but don't point to one disorder (depression: fatigue, sleep problems)
- Ideal basis = aetiopathogenesis and prognosis → mostly unknown → current systems are **descriptive** (neutral, no commitment to causes)
- Being descriptive makes it **standardisable** → structured interviews classify reliably

Categorical vs dimensional

- **Categorical** – illness differs **qualitatively** from health; clearly separable, non-overlapping categories, each with characteristic signs
    - used now; fits the clinical habit of 'ill or not ill'
- **Dimensional** – illness = **quantitative** variant of health on a continuum; disorder = value beyond an agreed threshold (like hypertension)
- Tension: natural borderlines between mental health and illness are unlikely → the categorical choice is pragmatic (→ section 5)

DSM and ICD

- Early ICD and DSM were based on **aetiological theories** (e.g. psychoanalysis)
- Turning point: **DSM-III** and **ICD-10** → descriptive, explicit criteria validated by studies
- DSM-5 gives **operational definitions** per disorder:
  - key and optional symptoms, minimum number required, often minimum duration
  - relevant physical conditions and psychosocial stressors
  - disability measured with WHODAS-2
  - codes taken from the ICD

Diagnosis

- Not standardisable; integrates history, mental state, physical exam, tests – and the **response to treatment** is itself diagnostic information
- Components of the integrated picture:
  - meaning of the symptoms for the patient and those close (the patient's narrative)
  - stage of development, severity, complications, extent
  - aetiological and pathogenetic factors – **predisposing, precipitating, perpetuating**; physical, mental, social; both illness-promoting and **protective**
  - factors influencing the course
  - which functions are dysregulated and how this affects functioning
  - risk of danger to self or others
  - factors influencing the expected treatment effect
- The patient's current behaviour can both **perpetuate** symptoms and help **solve** them

Example: the same patient classified as 'depressive disorder, recurrent, moderately severe, with anxious distress' is *diagnosed* as a third episode after incomplete recovery, with bipolar and depressive disorders in the family, father's

early death as a possible predisposing factor, relationship problems as perpetuating factor, avoidant/dependent traits, and past (not current) suicidal thinking.

Differential classification

The same symptom fits several disorders, so classification depends on the **pattern** of symptoms and on ruling out higher-ranked disorders first.

- Sadness as worked example:
  - sadness → depressive disorder
  - alternating with disinhibition and euphoria → **bipolar** disorder
  - meets depression criteria + anxiety → depressive disorder **with anxious distress**
  - severe anxiety is the main problem → **anxiety disorder**
- Hierarchy rule: rule out more important disorders with the same signs first
- But disorders coexist or follow each other: dementia + depression; schizophrenia + OCD; GAD preceding depression; alcohol use disorder secondary to social anxiety
- Uncertain → **provisional** classification + **differential classification** listing what cannot yet be confirmed or excluded

Algorithm for the main category

Output = the main category, reached by a stepwise rule-out that follows the hierarchy; it does not exclude coexisting disorders.

1. **Gate**: a disorder of cognitive, affective or conative functions **and** clinically relevant distress or impaired functioning – both required, otherwise no psychiatric disorder
2. Lowered consciousness, attention, orientation, memory, language, gnosis/praxis, executive or metacognitive disorder → **neurocognitive**
3. Hallucinations, delusions, disorganised speech or behaviour, catatonia → **psychosis spectrum**

4. Euphoria, irritability, increased energy, depressed mood, loss of interest, anhedonia → **bipolar / depressive**
5. Irrational anxiety, fear, panic → **anxiety**
6. Obsessions, compulsions or other repetitive behaviour → **obsessive-compulsive and related**
7. Emotional response to a major stressor → **trauma- and stressor-related**
8. Disturbed autobiographical memory, feelings of unreality → **dissociative**
9. Signs related to somatic symptoms → **somatic symptom, eating, sleep-wake, sexual dysfunction, gender dysphoria**
10. Uncontrolled harmful behaviour or harmful substance use → **impulse-control / paraphilic** or **substance-related / addictive**
11. Present since childhood → **neurodevelopmental**; since adolescence → **personality disorder**

5. RDoC (Research Domain Criteria)

RDoC is an NIMH research framework that **reverses** the diagnostic logic: instead of defining disorders by symptoms and then searching for their pathology, it studies fundamental psychological/biological functions across the full range from normal to abnormal, and treats psychopathology as **deviations** in those functions.

- It is **not** a diagnostic guide and does not replace DSM/ICD → it is a set of research principles
- = the dimensional approach (section 4), applied to functions rather than to syndromes

Why: the costs of symptom-based categories

Categories buy reliability and ease of use; RDoC targets what they cost research.

- **Heterogeneity** – many routes to the same diagnosis → two patients can share few or no symptoms → mechanisms differ within one category
- **Comorbidity** – patients often meet criteria for several disorders → more heterogeneity

    - studying only 'pure' single-diagnosis patients leaves dimensions **common across disorders** (transdiagnostic) unstudied
- **Arbitrary thresholds** – people just below the cutoff resemble those just above
    - a well/ill split hides how psychopathology emerges gradually, how risk factors work, and quantitative treatment outcomes
- **Late capture** – symptom-defined categories describe well-advanced, harder-to-treat problems
    - but trajectories shift gradually toward psychopathology → understanding the mechanisms across development could allow **pre-emptive** intervention

Structure of the framework

- **Domains** – six major types of neurobehavioural functioning:
    - arousal/regulatory · positive valence · negative valence · cognitive · social processes · sensorimotor
- **Constructs** – the psychological/biological dimensions within each domain, studied from normal to abnormal
- **Units of analysis** – classes of data to be measured and integrated: genetics, physiology, behaviour, self-report …
- Two cross-cutting factors:
    - **development** – functions change across the lifespan
    - **environment** – physical, cultural, social determinants of health
- **RDoC matrix** = domains/constructs × units of analysis
    - a heuristic of example research ideas, not a fixed or complete list

Aim

- Data on basic biological and behavioural processes → new assessment tools, **revisions of diagnostic systems**, prevention and treatment

DSM vs RDoC at a glance

	DSM / ICD	RDoC
Purpose	Clinical classification	Research framework
Unit	Disorder (syndrome)	Functional construct
Logic	Symptoms → category → seek pathology	Normal function → deviation → symptoms
Shape	Categorical, thresholds	Dimensional, normal-to-abnormal
Data	Mainly symptoms and signs	Multiple units of analysis, incl. biology
Comorbidity	Problem (heterogeneity)	Expected (shared dimensions)

6. Neuromodulation of neurons and synapses

Neuromodulators reconfigure 'hard-wired' circuits by changing the properties of neurons (ion channels → excitability) and synapses (strength and dynamics); their actions are divergent and convergent, often nonlinear, and span timescales from acute to long-term.

- Neuromodulators = small-molecule transmitters, biogenic amines, neuropeptides and others, released in modes **other than** classical fast synaptic transmission
- Targets: a circuit's neurons, its synapses, or its inputs → output adapts to behavioural context
- Two views that seem to conflict:
  - **global view** – one or a few modulators set brain states (sleep, arousal); presence of a modulator ≈ a behavioural state
  - **cellular view** – many modulators act on every neuron at once; excitability and synaptic efficacy are **always** under modulation → reconfiguration is a balance of synergistic and antagonistic pathways
  - not contradictory data → the difficulty of bridging cellular → circuit → behavioural levels

6.1 Canonical mechanism of ion-channel modulation

Output = the channel's phosphorylation state, which sets how it gates.

1. Modulator binds a **G-protein-coupled receptor** (GPCR) → receptor activates its G protein
2. G protein activates **second messengers**
3. Second messengers activate **kinases and phosphatases**
4. Kinases/phosphatases **phosphorylate or dephosphorylate** the channel protein
5. Different phosphorylation sites control different aspects of gating → changed availability, voltage- and time-dependence of activation/inactivation (sometimes unitary conductance)

- Other routes: direct gating by cyclic nucleotides or G proteins; tyrosine kinases → same theme: phosphorylation sites control gating
- Ubiquity: every channel is probably modulated at all times
  - even voltage-gated Na^+ channels, long thought unmodulated, have multiple phosphorylation sites → balanced input from many modulators
- A modulator can change a neuron by: **adding/subtracting** currents · changing the **relative magnitudes** across currents · changing **voltage/time dependence** of gating

6.2 Divergence and convergence

One modulator reaches many targets (divergence); many modulators reach one target (convergence) – so a neuron's state is a combination, not a single switch.

- **Divergence** (one modulator → many effects):
  - several channel types in one cell, via a shared pathway or different receptors
  - the same pathway can have **opposite** effects on different channels (one kinase pathway up-regulates one K^+ channel type, down-regulates another)
  - one modulator can act through **several** pathways that combine (dopamine on Ca^{2+} currents; μ-opioid on Na^+ currents via both PKA and PKC)

- **Convergence** (many modulators → one channel), same-sign effects are either:
  - **occluding** – simply additive, they share the mechanism
  - **synergistic** – more than additive
- Patterns across cell types (identified-neuron circuits):
  - **amines** (dopamine) – many currents in each cell, but **which** currents and the **sign** are cell-type-specific
  - **neuropeptides** – converge onto the **same** channel type, but each peptide acts on a **different subset of cells**
  - → predicting circuit effects requires mapping convergence/divergence both within one neuron and across neurons

6.3 Specificity despite diffuse release

Problem: modulators are released diffusely (or via the circulation) and second messengers are global, yet effects are specific to channel types and cell compartments.

1. Intracellular signalling is **restricted to local domains**
2. Channels and enzymes are organised into **local signalling complexes**, held by **anchoring proteins** = the specificity device
3. → only channels within that complex/compartment respond, even if the same channel type elsewhere, or another type nearby, could respond to the pathway
4. → local differences in gating properties within one cell

Evidence: in brainstem interneurons, dopamine (via PKC) inhibits T-type Ca^{2+} channels at the axon initial segment, altering action-potential initiation, but leaves Na^{+} channels there and T-type channels in soma/dendrites untouched – selective for both channel type and compartment.

6.4 Leak channels and ionotropic receptors

Excitability depends not only on voltage-gated channels but on **input resistance** and **resting potential**, which modulators set through leak and ligand-gated

channels.

- **Two-pore-domain K^+ channels** = major source of leak conductance
  - modulated by many intracellular pathways; convergent target of several modulators
  - **inhibition** → less leak → depolarisation, higher input resistance → **more excitable**
  - **activation** → hyperpolarisation → molecular target of anaesthetics
- **Ionotropic receptors acting as modulators**:
  - **ACh** – fast transmitter at the neuromuscular junction, but mostly a **neuromodulator in the CNS**, even through ionotropic receptors
  - **GABA_A** – besides fast inhibition, **extrasynaptic** receptors respond to low-level **tonic** GABA
  - **AMPA** – can have presynaptic modulatory actions
- **Axonal** modulation via ionotropic receptors alters action-potential generation and propagation

6.5 Excitability: straightforward vs nonlinear effects

- **Straightforward** – effects add up or synergise → firing goes up or down
  - e.g. 5-HT in motor neurons: ↓ outward currents + ↑ inward currents → more excitable
- **Nonlinear** – most neurons have nonlinear membrane behaviours: post-inhibitory rebound, plateau potentials, bistability, resonance, endogenous bursting
  - here 'more/less firing' is too simple → modulation **fine-tunes** these properties and can change membrane behaviour **qualitatively**
  - e.g. dopamine on one rhythmic neuron: increases some inward **and** outward currents, decreases others; its pre- and postsynaptic effects can oppose each other; it also changes synaptic strength and dynamics

6.6 Synapses: indirect vs direct modulation

- **Indirect** – via excitability:
  - presynaptic: change in action-potential **shape** → changed release
  - postsynaptic: e.g. ↑ voltage-gated inward currents → amplified EPSPs
- **Direct**:
  - presynaptic → **release probability**: Ca^{2+} influx, size of the reserve pool, active-zone proteins
  - postsynaptic → **receptor** expression and properties, independent of release
  - local feedback on release → retrograde messengers, autoreceptors
- **Meta-modulation** = modulation of modulation (e.g. nitric oxide altering the modulatory actions of glutamate or 5-HT)

6.7 Synaptic strength

- Several modulators can act on one synapse, presumably according to behavioural need
- Effects can be drastic: 5-HT can **silence** a synapse; dopamine can **unmask** a silent one
- Combined actions can be **more than additive**; one modulator can have **opposite** effects depending on network state
- Key lesson: a change in strength need **not** have an obvious network effect
  - dopamine strengthens an inhibitory synapse without changing the postsynaptic neuron's activity
  - neuropeptides greatly strengthen feedback onto pacemaker neurons with no change in network frequency – but it **stabilises** the oscillation (less variability)
- **Electrical synapses** (gap junctions) are also modulated (monoamines, NO)
  - but the functional effect may come mainly from changes in the neurons' **input conductance**, not the junction itself

6.8 Synaptic dynamics (short-term plasticity, STP)

- STP = synaptic efficacy changing during repetitive presynaptic activity (facilitation or depression)
    - → acts as **gain control**: strength becomes a function of presynaptic **frequency**
- Modulators alter STP and can even **switch its sign** (depression → facilitation)
- Strength (response to a **single** input) and STP (response to **sustained** input) can be modulated in **functionally opposite** directions → STP modulation matters as much as strength modulation for efficacy during network activity

6.9 Timescales: sensitisation of the Aplysia gill-and-siphon withdrawal reflex

Classic paradigm: serotonin converts a habituating sensory→motor synapse into a facilitated one, and longer exposure recruits progressively more durable mechanisms – from acute modulation to memory.

- Circuit: sensory neuron → motor neuron synapse mediates the reflex; it normally **habituates**
- Pairing touch with a noxious tail shock → serotonergic modulation → **sensitisation**

Short-term facilitation:

1. Tail shock → **5-HT** released onto the sensory neuron terminal
2. 5-HT → activates **PKA** (via cAMP signalling)
3. PKA → **reduces presynaptic K^+ currents**
4. Less K^+ repolarisation → **wider presynaptic action potential**
5. Wider spike → **more transmitter release** → stronger synapse

Intermediate-term facilitation (longer training ≈ prolonged 5-HT):

- presynaptic **PKC** + postsynaptic **Ca^{2+}-dependent** pathways → insertion of **AMPA-like receptors**

Long-term facilitation (long-term sensitisation of the behaviour):

1. Persistent PKA → phosphorylates **CREB** = the switch to long-term

2. CREB → changes **transcription**
3. Transcription → **new protein** expression
4. New proteins → **structural changes** at the synapse

6.10 Neuromodulator control of spike-timing-dependent plasticity (STDP)

Modulators gate long-term plasticity and set its sign through two opposing pathways – a push-pull rule.

- STDP normally: relative timing of pre- and postsynaptic spikes sets the direction of change (pre before post → LTP; post before pre → LTD)

1. Induction protocol enabling **NMDA-dependent** plasticity must be paired with the modulator → modulators act at induction
2. **PKA pathway** (e.g. β-adrenergic receptors) → promotes and **gates LTP**
3. **PLC pathway** (e.g. M1 muscarinic receptors) → promotes **LTD**
4. Each pathway **suppresses the other** = push-pull
5. Only PKA active → LTP regardless of timing; only PLC → LTD regardless of timing; **both** active → normal bidirectional, timing-dependent STDP

- The rule holds even at interneuron synapses where STDP relies on mGluR5 instead of NMDA receptors → independent of the underlying LTP/LTD mechanism

6.11 Long-term regulation of ion channels

Modulators do not only tweak channels acutely; their continued presence maintains which channels a neuron expresses and how their levels are co-regulated.

- Removing endogenous modulators → disrupts the **expression** and **co-regulation** of ion channels (and their mRNA), cell-type-dependently
  - via **transcription** and **translation**
- **Tonic** low-level dopamine → persistent increase of a transient K^+ current through translational regulation

- After spinal cord injury (loss of descending amines):
  - spinal interneurons become supersensitive to modulators
  - motor neurons recover persistent Ca^{2+} currents because amine receptors become **constitutively active**

6.12 From cell to circuit

- Circuit effects depend on the **type, location and timing** of modulator release
- A single modulator acts on many channels and on synaptic strength and dynamics → its effect on circuit output is not straightforward
- → needs new theoretical and computational models that integrate short- and long-term, multi-target actions